

Alpilla 600 MW Combined Cycle Power Plant

Progress Measurement and Reporting Procedure

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A	IFR	2026-12-14	First issue for Owner approval	EPCE	EPC project-control lead	EPC engineering manager

1. Purpose and scope

1.1 This procedure defines how progress of the Works is measured and reported: engineering documents, procurement, construction and commissioning, the Level 3 CPM schedule, S-curves, critical path and float, and the monthly progress report required by ER-17.02.

1.2 The method is applied by the engines of the project (progress_report, engineering_plan, release_control, schedule) on the data in the database; no progress figure is typed into a report by hand. This procedure is consistent with section 10 of ALP-EPC-00000-GE-PRC-0002 (release control and process gates) and with ALP-EPC-00000-GE-PLN-0001 (engineering execution plan).

1.3 It applies to Istanbul EPC, Imaginary Electric and all subsuppliers reporting progress to the project control function; subsuppliers report on the templates and dates set in their purchase orders.

2. Governing requirements and documents

ER-17.02: Level 3 CPM schedule (MS Project XML or Primavera P6) with monthly updates, S-curves and progress reports.

ER-17.01: Documents submitted electronically with the project numbering (consortium proposal adopted by the Owner in TQ-035: ALP-<ORG>-<KKS>-<DISC>-<TYPE>-<NNNN>) and a master document register; review period 14 days (21 days for the Basic Design Package).

2.1 Related documents: Level 3 Project Schedule ALP-EPC-00000-PC-SCH-0001, Advanced Work Packaging Execution Plan ALP-EPC-00000-GE-PLN-0002, Project Execution Plan ALP-EPC-00000-GE-PLN-0005, Document Numbering and Document Control Procedure ALP-EPC-00000-GE-PRC-0002.

3. Document progress measurement

3.1 Progress of the engineering and supplier documents is measured by weight. The weight of a document is its planned preparation hours (field 'weight' of the document record, in hours; for supplier documents the EPC review hours). A document without a weight counts as 1.

3.2 The earned percentage of a document is the highest percentage of the progress rule that corresponds to a revision issued on or before the data date: the rule for the revision purpose (IFR, IFA, IFC, AB), or the rule STARTED while the document is in progress and no revision has been issued. A cancelled document is excluded from the totals. The percentages are records (progress_rule) and are given in the table below.

Step	% earned	Remarks
STARTED	10	work started (first input received and preparation booked); ER-17.02, PC-PRC-0001 (to be agreed with the Owner)
IFR	40	issued for review / information; ER-17.02, PC-PRC-0001 (to be agreed with the Owner)
IFA	70	issued for approval (comments incorporated); ER-17.02, PC-PRC-0001 (to be agreed with the Owner)
IFC	90	issued for construction / final; ER-17.02, PC-PRC-0001 (to be agreed with the Owner)
AB	100	as-built or final record issue; ER-17.02, PC-PRC-0001 (to be agreed with the Owner)

3.3 The progress_rule records shall exist before the first monthly report: the progress_report engine stops with an error when they are missing. The percentages are proposed by project control, agreed with the Owner in the first progress meeting and changed only by a reviewed data change.

3.4 The document types and their review class, preparation time and progressive share are held in the doc_type records and are listed below for reference; the release maturity of each type is defined in section 4 of ALP-EPC-00000-GE-PRC-0002.

Type	Title	Review class
ARC	Architectural drawing	approval
BLK	Block / architecture diagram	approval
CAL	Calculation	review
CBL	Cable schedule and routing	information
CEM	Cause and effect matrix	approval
CMP	Commissioning procedure	approval
CNR	Control narrative / functional description	review
CVL	Civil drawing (grading, roads, drainage, marine)	approval
DBR	Design basis report	approval
DSH	Datasheet	approval
EFD	Equipment foundation drawings (vendor-certified loads)	approval
ELY	Electrical layout (earthing, lighting, trays, equipment)	review
FND	Foundation / underground drawing	approval
GAD	General arrangement drawing	review
HAZ	Safety study (HAZID, HAZOP, SIL, hazardous areas, fire, QRA)	approval
HMB	Heat and mass balance	approval
HMI	Operator display (HMI graphics) drawings	review
HUP	Instrument hook-up drawings	information
IOL	I/O list	review
ISO	Isometric drawing package	information
ITP	Inspection and test plan	approval
LAY	Equipment layout / arrangement	approval
LOG	Logic diagram	approval
LOP	Loop / connection diagram	information
LST	List / index (equipment, lines, valves, instruments, loads, tie-ins)	review
M3D	3D model review report (30 / 60 / 90 %)	review
MDB	Manufacturing record book / conformity certificates / turnover dossier	information
MDL	Master document list	information
MOS	Method statement / construction execution plan	review
MRQ	Material requisition	information
OMM	Operation and maintenance manual	review
PFD	Process flow diagram	approval
PID	Piping and instrumentation diagram	approval
PLN	Plan (execution, quality, HSE, commissioning, training)	approval
PLT	Plot plan	approval
PPL	Piping general arrangement / plans	review
PRC	Procedure / project standard	approval
RBR	Reinforcement drawings	review
RPT	Technical report	review
SCH	Schedule	review
SDS	System design description	review
SLD	Single line diagram	approval
SPC	Technical specification	approval
STL	Steel structure drawing	approval
STR	Pipe stress analysis report	review
STU	Engineering study	review
SUP	Pipe support drawings	information
TRP	Test report / test record / certificate	review
TST	Test procedure (FAT, SAT, site and performance tests)	approval
WIR	Schematic and wiring diagrams	information

3.5 The overall document progress is the sum over all documents of (earned percentage x weight) divided by the sum of the weights. Progress of a discipline, of a work package (EWP) or of a system is computed the same way over the documents of that group; the discipline breakdown is a table of the monthly report.

4. Planned progress, earned value and S-curve

4.1 The planned curve is built from the planned dates of each document (planned_ifr, planned_ifa, planned_ifc of the levelled workflow timeline of ALP-EPC-00000-GE-PLN-0001), each date credited with the percentage of its progress rule and the weight of the document; the earned curve is built from the issued revisions on their issue dates. Both are shown as weekly points as percentages of the total weight, the earned curve stops at the data date.

4.2 Earned value is reported in hours: earned hours = earned % x weight. The schedule performance index SPI = earned % / planned % at the data date; SPI and the difference planned - earned in percent points and in hours are reported for the total and for each discipline.

4.3 The planned curve is re-baselined only when the MDL revision or the levelled workflow timeline is formally revised (change of scope, staffing or key dates); the baseline of each revision is kept, and the monthly report states which baseline the planned curve follows. Documents late against the plan are listed with the days late and the reason (input, review, capacity).

4.4 The overall project progress is the weighted average of the progress of the activities of the Level 3 schedule with their budget hours as weight (field budget_hours of the activity record). Activity progress follows its deliverable documents (weighted by document weight) when it has deliverables, otherwise its recorded percent complete; an activity with an actual finish is 100 %.

5. Monthly progress report

5.1 The progress report is issued monthly to the Owner for each cut-off. The cut-off is the data date of the project record (project.data_date), set as the last calendar day of the month; only revisions issued and actuals recorded on or before the cut-off count. The report is issued within 5 working days after the cut-off.

5.2 The report contains: (1) summary and key dates with forecast and float; (2) overall progress planned versus earned, with the S-curve (engineering, procurement, construction, commissioning); (3) engineering: documents by status and discipline, documents issued in the month, late documents, documents in CHECK REQUIRED, review codes 3 and 4, open comments; (4) procurement: status of the requisitions, purchase orders and deliveries against the need dates; (5) construction: progress by CWP and work area, mechanical completion; (6) commissioning: systems by milestone; (7) critical and near-critical activities and float; (8) look-ahead of 3 months; (9) issues, risks and technical queries; (10) HSE and environmental data (compliance matrix ALP-EPC-00000-HS-RPT-0001).

5.3 The reports are produced by the engines from the database (R-001): progress_report gives the document S-curve, KPIs, discipline breakdown and late documents; engineering_plan gives the master document list, the AWP structure (CWA, CWP, EWP, PWP), the monthly load per discipline, EWP and PWP float and the coverage and benchmark checks; release_control gives the release plan, the relations, the impacts and the gate status (see ALP-EPC-00000-GE-PRC-0002 section 10); schedule gives the CPM schedule as Excel, HTML Gantt and MS Project XML for import into Primavera P6; document_register gives the register with the status of each document.

5.4 Before the cut-off the project control lead shall pull the shared branch, enter all issued revisions, actual dates and progress data through the database CLI with a reason, run python -m engine run --stale and python -m engine gate, check the warnings printed by the engines and resolve or explain each of them in the report. Reports are issued through python -m engine deliver.

5.5 A progress report figure that differs from the previous report without new data (baseline change, re-weighting, cancelled documents) is explained in the report. Issued revisions are frozen; a correction is a new revision and is shown as such.

6. Schedule levels, critical path and float

6.1 The schedule has three levels. Level 1: contract key dates and milestones (Notice to Proceed, back-feed energisation, GT first fire, first synchronisation in combined cycle, Taking-Over) as shown in ALP-EPC-00000-GE-PLN-0001. Level 2: phases and the work areas (CWA) with their construction work packages (CWP). Level 3: the CPM network of activities (activity records) with durations in working days, logic links (FS, SS, FF, SF with lags), constraints, deliverable documents and budget hours; it is issued as ALP-EPC-00000-PC-SCH-0001.

6.2 The Level 3 schedule is a CPM schedule calculated by the engine (planning core) on the working calendar of the project record; it is delivered as MS Project XML (P6 import) and Excel / HTML Gantt, updated every month at the data date: completed activities are fixed at their actual dates, activities in progress keep their actual start and remaining work starts at the data date, activities not started cannot start before the data date.

6.3 An activity is critical when its total float is zero or negative and it is not completed; near-critical activities have a float of less than 20 working days. Total float, free logic and constraints are reported for the key dates, and the longest path to Taking-Over is reported as a list of activities with float. Negative float is reported with its cause (capacity, logic, purchase order) before more staff or constraints are proposed.

6.4 Engineering timeline: the engineering documents are levelled against the capacity of each discipline (eng_resource records) in need-date order; EWP float is the time between the planned IFC of the EWP (plus its lead days) and the start of its CWP, PWP float is the time between the planned delivery on site of the material requisition and the CWP start. EWP and PWP with less than 20 working days float are listed as near-critical in ALP-EPC-00000-GE-PLN-0001 and in the monthly report.

6.5 Every schedule change is followed by python -m engine gate and python -m engine run --stale. Documents or activities dated beyond the project end are a data error and shall be corrected, not reported as a plan.

7. Document status and release control

7.1 The status of a document is the release status of ALP-EPC-00000-GE-PRC-0002 section 10: IFR (first issue), IFA (issued for approval), ACCEPTED (review code 1 or 2 on the latest issue; supplier documents when accepted by Istanbul EPC), IFC (issued for construction / final) and AB (as built). The status reported is that of the latest issued revision on the cut-off.

7.2 A document with an input revised after its issue is reported as CHECK REQUIRED; a review code 3 is reported as blocking the release of the documents that use it; documents blocked by their inputs are reported with the blocking inputs from the release plan (python -m engine doc status <document>).

7.3 Process gates (procurement, Owner acceptance, construction, commissioning, operations) are reported with their need date, planned date and gate float; every gate planned after its need date is reported as late and a technical query or recovery action is raised for it.

8. Procurement progress

Step	Credit	Remarks
Requisition issued (MRQ IFC)	5	Credit of the step; total 100 %. Weight of the package = order value [ER-17.02, DEC-EPCE-0009]
Purchase order awarded	15	[ER-17.02, DEC-EPCE-0009]
Supplier documents accepted	10	[ER-17.02, DEC-EPCE-0009]
Manufacturing (supplier progress reports, witnessed milestones)	40	[ER-17.02, DEC-EPCE-0009]
Factory acceptance test passed	15	[ER-17.02, DEC-EPCE-0009]
Shipped	5	[ER-17.02, DEC-EPCE-0009]
Delivered on site, inspected	10	[ER-17.02, DEC-EPCE-0009]

8.1 Procurement progress of a package is the sum of the credits of the steps completed (partial credit for manufacturing according to the supplier's certified progress); the progress of the procurement phase is the average of the packages weighted by the order value (or the budget hours when no order value is known). The steps follow the procurement gates of section 10 of ALP-EPC-00000-GE-PRC-0002 (enquiry, purchase order, manufacturing release, factory acceptance test, shipment). Delivery on site is compared with the need date of the first CWP start and the float is reported.

9. Construction and commissioning progress

Step	Credit	Remarks
CWP released (EWP complete, materials on site or confirmed)	5	Credit of the step of a CWP; total 100 % [ER-17.02, DEC-EPCE-0012]
Installation by quantity (installed / planned quantities of the CWP)	75	[ER-17.02, DEC-EPCE-0012]
Inspection and test plan records closed	10	[ER-17.02, DEC-EPCE-0012]
Mechanical completion certificate	10	[ER-17.02, DEC-EPCE-0012]

9.1 Construction progress is measured by quantities: the installed quantity of each CWP (quantities field of the CWP: concrete, piles, steel, equipment, pipe, cable, instruments) is recorded per IWP each week and credited against the planned quantity; where quantities are not defined the work is measured by the milestone steps above. The progress of a work area is the average of its CWP weighted by budget hours; the CWP release follows the EWP and PWP conditions of ALP-EPC-00000-GE-PLN-0002.

9.2 Commissioning progress is measured by systems: pre-commissioning, mechanical completion, cold commissioning, hot commissioning and system completion per the commissioning gates; the progress of the commissioning phase is the average of the systems weighted by the budget hours of their commissioning activities. Mechanical completion of the CWPs drives the key dates of Level 1, and the contractual tests (Tests on Completion) are shown as activities with their planned dates.

9.3 Payment milestones of the contract are reported with their trigger and status; a milestone is claimed only when its trigger is met and evidenced by the document or certificate named in the contract.

10. Key performance indicators

KPI	Threshold / reporting	Remarks
Schedule performance index, engineering (earned / planned at the data date)	alert below 0.95, critical below 0.90	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Documents issued IFC against plan (count and weight)	difference to planned IFC in the month	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Late documents (planned date before the data date, not issued)	count and days late	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]

KPI	Threshold / reporting	Remarks
Documents CHECK REQUIRED or with review code 3	count, all to be cleared within 10 working days	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Process gates planned after need date	count, all reported with recovery action	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
EWP and PWP with float below 20 working days	count and list	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Activities with negative float	count, list and cause	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Engineering hours used against the levelled capacity of each discipline	FTE used against the eng_resource capacity	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Milestones and key dates: forecast against contract date	working days of float	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Open technical queries and clarifications	count, ageing	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]
Procurement: packages delivered on time against need date	percent of packages	[ER-17.02, ALP-EPC-00000-GE-PLN-0001]

10.1 KPI thresholds are set by project control for the first issue and are reviewed with the Owner after three monthly reports. An indicator in the alert range is explained in the report with the cause and the corrective action.